# Graphical Linear Algebra: From Matrices to Lagrangian Relations

N-20210621

A matrix is an economical way to describe a linear map, but matrix multiplication hides a remarkably simple fact. In a network, a signal can reach an output along several paths. Each path contributes the product of its weights, while the paths ending at the same output have their contributions added. Thus

$$(AB)_{ij} = \sum_r A_{ir} B_{rj}$$

is already a wiring instruction: the repeated index $r$ marks a collection of intermediate wires that are connected and then hidden.

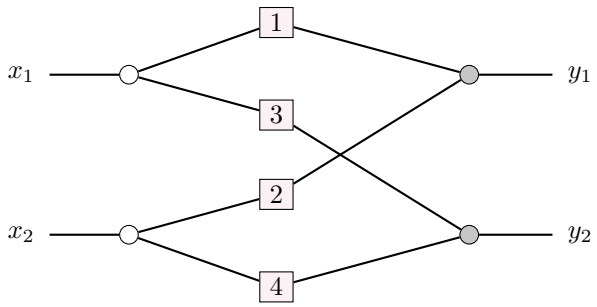
Could one perform the algebra with the wires themselves, transforming one network into another by small local moves? The question becomes even more interesting when a network specifies an equation which has several solutions, or no solution at all. We shall begin with the matrix operations, then allow equations to be read in either direction, and finally ask which such relations preserve a symplectic form. The resulting path leads from the PROP $\mathbf{cb}_k$ to $\mathbf{ih}_k$, and then to Lagrangian relations.[1]

## §1.1 A linear map on four paths

Start with

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \qquad \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

The two outputs can be assembled with three kinds of operation: copy each input, multiply each copy by a weight, and add all contributions to an output. We draw white circles for copying and grey circles for addition. In every diagram of this note, *inputs are on the left and outputs on the right.*




A crossing without a node leaves the two wires separate.

Figure 1.1: The matrix $A$ expressed as a weighted network. Reading from left to right gives $y_1 = x_1 + 2x_2$ and $y_2 = 3x_1 + 4x_2$.

[1] The first part follows Paweł Sobociński, *Graphical Linear Algebra*, especially Episodes 7–30. Our white-copy/grey-add convention follows the ZX-style colouring used by Comfort and Kissinger. The complete matrix and relation presentations are developed in Bonchi, Sobociński and Zanasi, *Interacting Hopf Algebras* (2017).

A diagram of this kind is a mathematical expression: connecting two diagrams ought to mean composing the transformations. If we want to *prove* equalities by moving its nodes, however, we must specify both the types of the diagrams and the equations that justify moving them.

## §1.2 Why a number of wires is an object

Fix a commutative ring $k$ for the elementary calculations; a field will be required later for some geometric results. The category $\mathsf{Mat}_k$ uses natural numbers $0, 1, 2, \ldots$ as objects. Its arrows are

$$\mathsf{Mat}_k(n, m) = \{m \times n \text{ matrices over } k\}.$$

The number $n$ stands for a free module $k^n$. An arrow $A : n \to m$ is therefore a linear map $k^n \to k^m$.

There are two ways to combine arrows. For $A : n \to m$ and $B : m \to p$, connecting the output of $A$ to the input of $B$ means $B \circ A = BA : n \to p$. For $C : r \to s$, putting $A$ and $C$ next to each other means the block diagonal matrix

$$A \oplus C = \begin{pmatrix} A & 0 \\ 0 & C \end{pmatrix} : n + r \to m + s.$$

The switch of two adjacent wires is a permutation matrix $\sigma : 2 \to 2$, and permutations of larger collections of wires are built from such switches.

A *PROP* is a strict symmetric monoidal category with object set $\mathbb{N}$ and monoidal product on objects given by addition. Here its monoidal product of arrows is the *direct sum*. There is an important distinction:

$$\text{two parallel wires} \quad \longleftrightarrow \quad k \oplus k,$$

which has rank 2. It does not mean $k \otimes_k k$, which has rank 1. Confusing these operations would completely alter the meaning of graphical copying.

## §1.3 Five operations and their exact types

The graphical theory $\mathbf{cb}_k$ has four basic structural operations, plus a scalar operation for every $a \in k$. These are maps of finite free modules:

| operation | type | value on inputs |
|---|---|---|
| $\Delta$ (copy) | $1 \to 2$ | $x \mapsto (x, x)$ |
| $\epsilon$ (discard) | $1 \to 0$ | $x \mapsto 0 \in k^0$ |
| $\mu$ (add) | $2 \to 1$ | $(x, y) \mapsto x + y$ |
| $\eta$ (zero) | $0 \to 1$ | $0 \in k^0 \mapsto 0$ |
| $[a]$ (scalar) | $1 \to 1$ | $x \mapsto ax$ |

The zero-dimensional module $k^0$ has only one element. In ordinary matrices these generators are

$$\Delta = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \epsilon = 0_{0\times 1}, \quad \mu = \begin{pmatrix} 1 & 1 \end{pmatrix}, \quad \eta = 0_{1\times 0}, \quad [a] = \begin{pmatrix} a \end{pmatrix}.$$

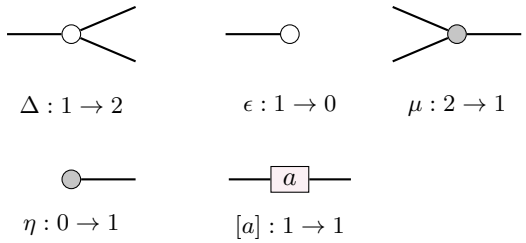



Figure 1.2: Five elementary diagrams. Their boundary types are part of the notation, and the two colours carry different operations.

The copying operation is genuinely linear:

$$\Delta(x+y) = (x+y, x+y) = \Delta(x) + \Delta(y)$$

where addition on the right is componentwise. By comparison, the map $v \mapsto v \otimes v$ is generally nonlinear, since $(v+w)\otimes(v+w)$ contains cross terms. We can copy a classical value into a direct sum without assuming that arbitrary vectors may be copied in a tensor product.

The types already catch elementary mistakes. The composite $\eta\epsilon : 1 \to 1$ is the zero map $x \mapsto 0$; reversing the order gives $\epsilon\eta : 0 \to 0$, which has an entirely different domain and codomain.

## §1.4 The equations behind moving the nodes

The grey addition operation is associative, commutative and has zero as its unit. Diagrammatically this says

$$\mu(\mu \oplus \mathrm{id}) = \mu(\mathrm{id} \oplus \mu), \qquad \mu\sigma = \mu, \qquad \mu(\eta \oplus \mathrm{id}) = \mathrm{id}.$$

Copy and discard obey the equations with all arrows reversed:

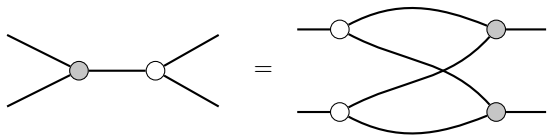
$$(\Delta \oplus \mathrm{id})\Delta = (\mathrm{id} \oplus \Delta)\Delta, \quad \sigma\Delta = \Delta, \quad (\epsilon \oplus \mathrm{id})\Delta = \mathrm{id}.$$

We call the former a commutative monoid and the latter a cocommutative comonoid. The equations authorize local substitution *inside* larger diagrams: adding three values in two bracketings really is the same map.

The first interesting law concerns the interaction of the two colours. If we add $x$ and $y$, then copy the result, we obtain $(x+y, x+y)$. Alternatively we may copy both inputs, exchange the middle two resulting wires and add corresponding pairs:

$$\boxed{\Delta\mu = (\mu \oplus \mu)(\mathrm{id} \oplus \sigma \oplus \mathrm{id})(\Delta \oplus \Delta).}$$

Both sides have type $2 \to 2$; the switch is essential for the types and the intended pairing.




add, then copy

copy, exchange, then add

Figure 1.3: The bialgebra law. On either side the output is $(x+y, x+y)$; crossing wires without a node remain distinct.

In ordinary matrices the left side is

$$\begin{pmatrix}1\\1\end{pmatrix}\begin{pmatrix}1 & 1\end{pmatrix} = \begin{pmatrix}1 & 1\\1 & 1\end{pmatrix}.$$

The right side gives the same matrix by multiplying the appropriate block and permutation matrices. There are also laws saying that the zero can be copied, that sums can be discarded, and that scalar nodes cooperate with the two structures.

Scalars are particularly instructive. Two beads in series multiply, while a copy followed by two weighted branches and an addition produces their sum:

$$[b]\,[a] = [ba], \qquad [1] = \mathrm{id}_1, \qquad \boxed{[a+b] = \mu([a]\oplus[b])\Delta}, \qquad [0] = \eta\epsilon.$$

These identities explain how several paths can collapse to a single scalar. They are examples of the full equations for a bicommutative bialgebra with the scalar family; our displayed selection is meant to explain the mechanisms, not silently replace the complete presentation.

## §1.5 A normal form for every matrix

For an arbitrary $m \times n$ matrix $A = (a_{ij})$, there is a systematic recipe. Copy each input $x_j$ into $m$ branches, multiply its $i$-th branch by $a_{ij}$, permute branches so that the $n$ contributions to output $i$ sit together, and add them. The result is the map

$$y_i = \sum_{j=1}^{n} a_{ij}x_j.$$

Write $\Delta_m : 1 \to m$ for repeated copying, $\mu_n : n \to 1$ for repeated addition, and $P_{m,n}$ for the wire permutation changing input-major to output-major order. Then the diagram can be written as a typed composite:

$$\mathcal{D}(A) = \left(\bigoplus_{i=1}^{m}\mu_n\right) P_{m,n} \left(\bigoplus_{j=1}^{n}\bigoplus_{i=1}^{m}[a_{ij}]\right) \left(\bigoplus_{j=1}^{n}\Delta_m\right).$$

Its intermediate types are

$$n \longrightarrow mn \longrightarrow mn \longrightarrow mn \longrightarrow m.$$

The conventions $\Delta_0 = \epsilon$ and $\mu_0 = \eta$ accommodate empty boundaries.

This construction shows that *all matrices have diagrams*. The converse is a subtler theorem: the equations suffice to prove equality of any two diagrams which denote the same matrix. A normalization argument repeatedly moves copying past addition using the bialgebra law, groups wires using associativity and permutation laws, and combines repeated weighted paths using the scalar laws. Eventually the diagram takes the copy–scalar–add form corresponding to its matrix entries. A complete proof must show that each normalization step is licensed by the full list of equations.

**Theorem 1.5.1** (The matrix presentation)

For a commutative ring $k$, the PROP $\mathbf{cb}_k$ of bicommutative bialgebra diagrams with its complete scalar equations presents the PROP $\mathsf{Mat}_k$:

$$\boxed{\mathbf{cb}_k \cong \mathsf{Mat}_k.}$$

In particular, graphical provability and equality of denoted matrices coincide.

The essential distinction is between *expressiveness*, which the construction $\mathcal{D}(A)$ establishes directly, and *completeness*, which requires the full equational normal-form argument. An attractive circuit drawing alone gives neither a proof of faithfulness nor an algorithm for rewriting every equivalent diagram.

## §1.6 Matrix multiplication counts weighted paths

Let

$$B = \begin{pmatrix} 3 & 0 \\ 4 & 1 \end{pmatrix}, \qquad A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

Passing signals first through $B$ and then through $A$ means multiplying in the order

$$AB = \begin{pmatrix} 11 & 2 \\ 4 & 1 \end{pmatrix}.$$

The coefficient 11 is a sum of contributions from two paths: $1 \cdot 3$ through the first intermediate wire and $2 \cdot 4$ through the second.

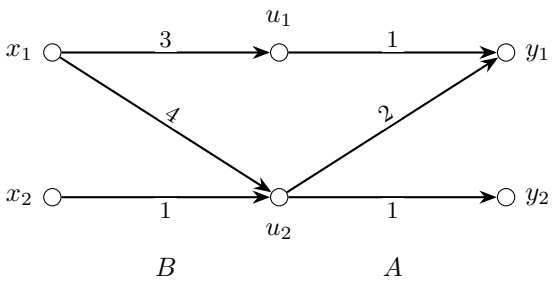



Figure 1.4: The path-sum rule for two connected linear networks. Zero-weight edges have been omitted; each path contributes a product and parallel paths contribute a sum.

The path-sum rule follows by induction for finite acyclic networks built from copy, scalars and addition. Composition concatenates paths, while addition merges their contributions:

$$(AB)_{ij} = \sum_{r} A_{ir} B_{rj}.$$

It is important not to extrapolate this formula to an arbitrary directed feedback loop by declaring an infinite sum of paths. Feedback introduces simultaneous equations; without invertibility or convergence it may fail to determine a function. A language of *relations* will handle those cases without pretending that every feedback network has a unique solution.

## §1.7 Negative coefficients and the Hopf antipode

The scalar $[-1]$ deserves attention because it permits cancellation. Write $S = [-1] : 1 \to 1$. Then

$$\boxed{\mu(\mathrm{id} \oplus S)\Delta = \eta\epsilon : 1 \to 1.}$$

Both sides send $x$ to zero: copy it, negate one copy, and add. In Hopf-algebra terminology $S$ is the *antipode*.

There is a genuine hierarchy here. The copy/add calculus with natural-number coefficients describes nonnegative integer matrices; allowing negative coefficients reaches integer matrices; admitting scalar beads indexed by a ring $k$ yields matrices over that ring. Cancellation cannot be used in an equational system whose available scalars do not admit additive inverses. The antipode is the structural form of subtraction, and it will also explain sign conventions when arrows are reversed.

## §1.8 What happens when an arrow is reversed?

Consider a perfectly ordinary equation:

$$z = x + y.$$

When $x, y$ are the inputs and $z$ the output, this is the linear map $\mu : k^2 \to k$. But suppose we are given $z$ and asked to recover $(x, y)$. Over a field, there are many pairs satisfying the equation:

$$(x, y) = (t, z - t), \qquad t \in k.$$

The same constraint makes sense in both directions, although it is a function in only one of them.

A *linear relation* $R : n \to m$ is a linear subspace

$$R \subseteq k^n \oplus k^m.$$

We write $x \, R \, y$ when $(x, y) \in R$. An input may be related to no outputs, one output, or many outputs. A linear map $A : k^n \to k^m$ determines the relation

$$\Gamma_A = \{(x, Ax) : x \in k^n\}.$$

The *converse* of a relation reverses its two boundaries:

$$R^\dagger = \{(y, x) : (x, y) \in R\}.$$

This operation always exists. When $R = \Gamma_A$, however, its converse need not be the graph of a function.

For our two basic generators, their converses behave quite differently:

$$\mu^\dagger = \{(z, (x, y)) : x + y = z\},$$
$$\Delta^\dagger = \{((x, y), z) : x = y = z\}.$$

Thus the converse of addition splits a given value into *any* two summands, while the converse of copying tests that two values agree. Merely observing that both $\mu^\dagger$ and $\Delta$ have type $1 \to 2$ does not make them equal.

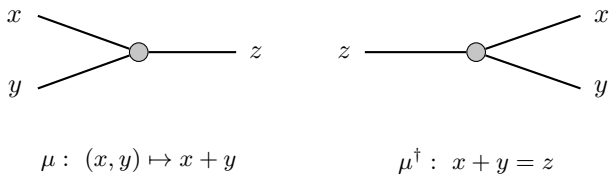



Figure 1.5: The addition constraint oriented in opposite directions. A grey node with two outputs means the *converse relation* of addition, and has no deterministic output.

This is a useful warning about diagrams: reflection is an operation on *relations*, and its semantics must be stated. It is generally neither matrix inversion nor matrix transposition.

## §1.9 Connecting constraints means eliminating variables

Let $R : n \to m$ and $S : m \to p$ be linear relations. Connecting the output of $R$ to the input of $S$ means

$$\boxed{S \circ R = \{(x, z) : \exists y \in k^m,\ (x, y) \in R,\ (y, z) \in S\}.}$$

The intermediate wire is an existentially quantified variable. Relational composition therefore performs a familiar operation: *eliminating hidden variables in a system of equations*.

For example, let

$$R = \{(x, (u, v)) : u = x,\ v = x\}, \quad S = \{((u, v), z) : z = u + 2v\}.$$

The composite eliminates $u, v$ and leaves the graph $z = 3x$. But if $R$ says $u + v = x$, then for each $x$ there are many allowed pairs $(u, v)$; this is nevertheless a perfectly good linear relation.

Here is a powerful normal form, derived from ordinary linear algebra.

**Proposition 1.9.1** (Every linear relation is a span of two maps)

For each linear relation $R \subseteq k^n \oplus k^m$ over a field, there are linear maps $X : k^r \to k^n$, $Y : k^r \to k^m$ with

$$\boxed{R = \Gamma_Y \circ \Gamma_X^\dagger.}$$

*Proof.* Take a basis of $R$, let $r = \dim R$, and collect the first and second components of the basis vectors as the columns of $X$ and $Y$. The set of all pairs generated by this basis is

$$R = \{(Xt, Yt) : t \in k^r\}.$$

But $(x, y) \in \Gamma_Y \circ \Gamma_X^\dagger$ means precisely that there exists $t \in k^r$ with $x = Xt$ and $y = Yt$. □

The same relation may be represented by *equations* rather than by parameters. Choose a matrix $C$ with kernel $R$, and write

$$R = \{(x, y) : C \begin{pmatrix} x \\ y \end{pmatrix} = 0\}.$$

The span representation describes solutions; the kernel representation describes constraints. Gaussian elimination translates one into the other.

These elementary facts already explain why one wants to add reflected copies of the matrix generators to a diagrammatic language. They generate every relation, but generation alone does not settle whether two diagrams expressing the same subspace can be transformed into each other by the rules.

**Theorem 1.9.2** (Interacting Hopf algebras)

For a field $k$, the PROP $\mathbf{ih}_k$ formed from forward and reversed matrix generators with the complete interacting-Hopf equations presents the PROP of linear relations:

$$\boxed{\mathbf{ih}_k \cong \mathsf{LinRel}_k.}$$

The completeness theorem must account for redundant parameters in a span, different matrices cutting out the same kernel, and all the ways these choices interact during composition. The equations implement those changes of presentation graphically. A proof that all relations have *some* diagram, such as the proposition above, is only half the story.

## §1.10 The zero scalar is not invertible, but it has a converse

This distinction between a function and a relation explains the provocative expression *division by zero* in the graphical linear algebra series.

Let $[a]$ denote multiplication by $a$. Reversing its graph gives

$$[a]^\dagger = \{(y, x) : y = ax\}.$$

When $a \neq 0$ in a field, this is the ordinary function $y \mapsto a^{-1}y$. When $a = 0$, instead,

$$\boxed{[0]^\dagger = \{(0, x) : x \in k\}.}$$

It accepts the input 0, and there it allows *every* output. At a nonzero input it has no solution. There is no numerical object $1/0$ hiding in the picture; the reversed relation remains meaningful because equations do not need to have unique solutions.

This becomes especially useful in feedback. Suppose a value satisfies the loop equation

$$x = u + rx,$$

where $u$ is an external input. The network denotes

$$F_r = \{(u, x) : (1 - r)x = u\}.$$

For $r \neq 1$, it is the ordinary graph $x = (1 - r)^{-1}u$. For $r = 1$, it becomes the relation $\{(0, x) : x \in k\}$. The loop is then inconsistent for $u \neq 0$ and underdetermined for $u = 0$.

A vector-valued version makes the link to classical linear algebra even clearer:

$$F_M = \{(u, x) : (I - M)x = u\}.$$

If $I - M$ is invertible, the relation is a function. Otherwise there can be no solutions when $u \notin \mathrm{im}(I - M)$, and an affine family of solutions indexed by $\ker(I - M)$ when $u$ lies in the image. A relational circuit retains both possibilities without imposing invertibility by fiat.

## §1.11 Two Frobenius structures and the spider rule

Some very short calculations explain why interacting graphical calculi admit dramatic simplifications. Relational converse gives

$$\boxed{\Delta^\dagger\Delta = \mathrm{id}_1, \qquad \mu\mu^\dagger = \mathrm{id}_1.}$$

To verify the first, copy $x$ into $(x, x)$, then apply the equality test; the only possible result is $x$. For the second, split $z$ into any two summands and add them back; every possible route returns $z$.

Reversing the order does *not* give identities:

$$\Delta\Delta^\dagger = \{((a,b),(c,d)) : a = b = c = d\},$$
$$\mu^\dagger\mu = \{((a,b),(c,d)) : a + b = c + d\}.$$

The second relation preserves only the total sum, while the first demands equality of all four entries. Their difference is a concrete reason why arbitrary nodes cannot be cancelled as if they were isomorphisms.

The copy operation with its converse, and the addition operation with its converse, each satisfy the equations of a *special commutative Frobenius algebra* in $\mathsf{LinRel}_k$. The corresponding spider theorem says that a connected network consisting exclusively of one such family of nodes depends only on the number of its exposed inputs and outputs. For the equality family, a connected spider is simply the relation that all incident variables have the same value. A tree of equality tests and copies can therefore be contracted into one node.

This theorem is a powerful *local* rewriting principle. Mixing the two colours introduces additional interaction equations; one cannot fuse a copy spider with an add spider just because their pictures look alike.

## §1.12 The missing structure: phase space

The graphical language is now expressive enough to handle general linear constraints, including reversed arrows and feedback. A further problem arises in mechanics and in certain electrical and quantum systems: two kinds of variables travel together. A configuration variable $q$ is paired with a momentum-like or covector variable $p$.

Let $V = k^n$ over a field $k$, and write its cotangent phase space in coordinates as

$$T^*V \cong k^n_q \oplus k^n_p.$$

The canonical symplectic form is the alternating, nondegenerate bilinear form

$$\boxed{\omega((q,p),(q',p')) = q^\mathsf{T}p' - p^\mathsf{T}q'.}$$

Its matrix is

$$J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

The word *alternating*, meaning $\omega(v,v) = 0$, is important in characteristic 2, where an equation involving only skew-symmetry can be misleading.

For a subspace $L \subseteq T^*V$, its symplectic orthogonal is

$$L^\omega = \{w : \omega(w,\ell) = 0 \text{ for all } \ell \in L\}.$$

We call $L$ *Lagrangian* when $L = L^\omega$. Because $\omega$ is nondegenerate, a subspace is Lagrangian precisely when $\omega$ vanishes on it and its dimension equals $n$.

What does this condition mean for a familiar matrix? Given $A \in k^{n\times n}$, consider the graph

$$L_A = \{(q,p) : p = Aq\}.$$

It has dimension $n$, and for its two generic points we calculate

$$\begin{aligned}\omega((q,Aq),(q',Aq')) &= q^\mathsf{T}Aq' - (Aq)^\mathsf{T}q' \\ &= q^\mathsf{T}(A - A^\mathsf{T})q'.\end{aligned}$$

Thus

$$\boxed{L_A \text{ is Lagrangian} \iff A = A^{\mathsf{T}}.}$$

We have found matrix symmetry again, now as the necessary and sufficient condition for a graph in phase space to be compatible with the symplectic pairing.

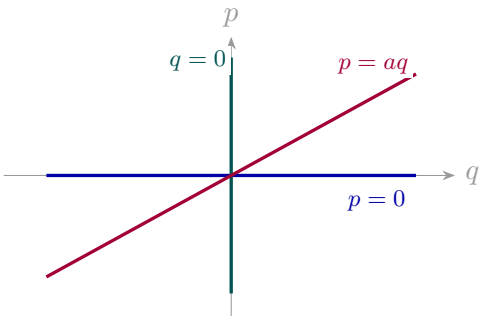



Figure 1.6: Lagrangian lines in the real phase plane. The two axes will arise from doubling zero and full subspaces; a nonhorizontal, nonvertical line exposes the need for an additional generator.

In one degree of freedom, every line through the origin is isotropic and therefore Lagrangian. In two degrees of freedom, the symmetric matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ gives the Lagrangian plane $p_1 = q_2,\ p_2 = q_1$, while the nonsymmetric matrix $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ does not.

## §1.13 A matrix on positions determines a relation on momenta

A linear map $A : k^n \to k^m$ sends a position $q$ to $Q = Aq$. But what should happen to a momentum-like covector? Its natural transformation runs in the other direction:

$$p = A^{\mathsf{T}} P.$$

Together the two equations define the *cotangent-lift relation*

$$\boxed{\mathcal{L}_A = \{((q,p),(Q,P)) : Q = Aq,\ p = A^{\mathsf{T}} P\}.}$$

The construction is a relation even when $A$ is singular or rectangular. It has as many solutions as its freely chosen parameters $q \in k^n$ and $P \in k^m$ permit.

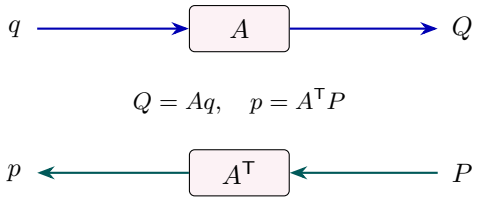



Figure 1.7: A position map $A$ and its transpose on covectors form a Lagrangian relation. The opposing arrows indicate how to solve the defining constraints, without asserting that the relation is a function.

The ambient space for a relation between phase spaces is

$$\overline{T^*k^n} \oplus T^*k^m,$$

where the bar reverses the symplectic form on the input. We can verify explicitly that $\mathcal{L}_A$ is Lagrangian. It has dimension $n+m$, half the dimension $2(n+m)$ of the ambient space. Its symplectic pairing is zero:

$$\begin{aligned}&\omega_m((Aq,P),(Aq',P')) - \omega_n((q,A^\mathsf{T}P),(q',A^\mathsf{T}P'))\\ &= q^\mathsf{T}A^\mathsf{T}P' - P^\mathsf{T}Aq' - q^\mathsf{T}A^\mathsf{T}P' + P^\mathsf{T}Aq' = 0.\end{aligned}$$

Furthermore, for $B : k^m \to k^r$, interconnecting these lifted relations gives

$$Q = Aq, \quad Z = BQ, \quad p = A^\mathsf{T}P, \quad P = B^\mathsf{T}R.$$

Eliminating $Q, P$ yields $Z = (BA)q,\ p = (BA)^\mathsf{T}R$, hence

$$\boxed{\mathcal{L}_B \circ \mathcal{L}_A = \mathcal{L}_{BA}.}$$

The transpose arises from the symplectic pairing, and the reversed momentum flow makes the composition work without requiring either $A$ or $B$ to be invertible.

## §1.14 Doubling a linear relation with its annihilator

We can remove the assumption that our starting object was the graph of a matrix. Let $U = k^n,\ V = k^m$ and let $R \subseteq U \oplus V$ be any linear relation. Define its annihilator

$$R^\perp = \{(\alpha,\beta) \in U^* \oplus V^* : \alpha(u) + \beta(v) = 0 \text{ for all } (u,v) \in R\}.$$

Now construct

$$\boxed{\mathcal{L}_R = \left\{((u,p_U),(v,p_V)) : (u,v) \in R, \quad (-p_U,p_V) \in R^\perp\right\}.}$$

The minus sign is essential: the symplectic form on the *input* phase space has been reversed. Without it, even the matrix case would not give the cotangent-lift equation $p_U = A^\mathsf{T}p_V$.

Here is a direct proof that the construction is Lagrangian. Since

$$\dim R + \dim R^\perp = n + m,$$

the doubled relation has dimension $n+m$, half that of its ambient symplectic space. Let $(r,s)$ and $(r',s')$ represent two of its elements with $r, r' \in R$ and $s, s' \in R^\perp$, using the sign convention above. The ambient symplectic pairing is

$$s'(r) - s(r') = 0$$

because the covectors annihilate $R$. The relation is isotropic and half-dimensional, hence Lagrangian.

For $R = \Gamma_A$, a covector $(\alpha,\beta)$ annihilates its graph precisely when $\alpha = -A^\mathsf{T}\beta$. Thus

$$(-p_U,p_V) \in \Gamma_A^\perp \quad \Longleftrightarrow \quad p_U = A^\mathsf{T}p_V.$$

The earlier cotangent lift was the graph case of this more general *doubling* construction.

One also needs to know that doubling respects composition and tensoring. For finite-dimensional vector spaces over a field, orthogonal-complement identities under linear projection and intersection give

$$\mathcal{L}_{S\circ R} = \mathcal{L}_S \circ \mathcal{L}_R.$$

Together with the natural reordering of position and momentum summands, this makes doubling a faithful strong symmetric monoidal functor

$$\mathsf{LinRel}_k \longrightarrow \mathsf{LagRel}_k.$$

Faithfulness is immediate from the position component: one can recover $R$ by projecting $\mathcal{L}_R$ onto the original position variables. The compositional statement is a verified theorem, and rests on the annihilator calculus for finite-dimensional relations.[2]

## §1.15 A Lagrangian relation that doubling cannot produce

The doubling functor is faithful, but it is not *full*. The simplest obstruction appears in just one degree of freedom.

A linear relation $R : 0 \to 1$ is a linear subspace of the one-dimensional space $k$. There are only two choices: $R = 0$ or $R = k$. Their doubles in $T^*k$ are

$$\mathcal{L}_0 = \{(q, p) : q = 0\}, \qquad \mathcal{L}_k = \{(q, p) : p = 0\}.$$

Yet for any nonzero $a$, the line

$$L_a = \{(q, p) : p = aq\}$$

is Lagrangian, as we established earlier, and is neither axis. Therefore it is *not* in the image of doubling.

How can we create it? Consider the shear

$$S_a : (q, p) \longmapsto (q, p + aq).$$

For two vectors,

$$\begin{aligned} \omega(S_a(q, p), S_a(q', p')) &= q(p' + aq') - (p + aq)q' \\ &= qp' - pq'. \end{aligned}$$

Thus $S_a$ is symplectic, and it sends the horizontal axis $p = 0$ to $p = aq$. Another symplectic operation is the Fourier-type rotation

$$F : (q, p) \longmapsto (-p, q).$$

It exchanges the two coordinate directions, with the sign necessary for preservation of the symplectic form.

These are concrete new operations which the purely doubled subcategory lacks. Comfort and Kissinger prove that, over any field, doubled linear relations together with the indicated Fourier and shear generators generate *all* linear Lagrangian relations. Their constructive proof uses symplectic Gaussian-elimination operations on a matrix representation of a Lagrangian subspace.

The distinction between generation and a stand-alone equation system matters here, just as it did for matrices. The paper gives a complete description via the already complete graphical calculus of linear relations, supplemented by appropriate generators. It does not provide a small, independent complete list of diagrammatic equations written solely in its chosen Lagrangian generators; this is explicitly left as a further direction.

[2] Cole Comfort and Aleks Kissinger, *A Graphical Calculus for Lagrangian Relations*, 2022, especially Lemma 2.7 and Theorems 3.1–3.2. Their paper also states the matrix and linear-relation presentations in Section 1.

## §1.16 Affine shifts and a carefully scoped quantum connection

Every linear relation is a subspace, so it contains the zero vector. To describe $p = aq + b$ with $b \neq 0$, we must allow an *affine* Lagrangian subspace, or a translate of a linear Lagrangian subspace. The translations

$$(q, p) \longmapsto (q + b, p + c)$$

preserve the symplectic form on displacement vectors. One extra affine-shift generator supplies the additional expressive power needed for affine Lagrangian relations.

A major application now becomes intelligible. Over an odd prime field $\mathbb{F}_p$, the paper proves a symmetric monoidal equivalence between affine Lagrangian relations and the corresponding $p$-dimensional qudit stabilizer process theory, *modulo invertible scalars*. The restriction to odd primes is substantive; the statement does not claim an equivalence with all quantum computations or an identical theorem for qubits.

Another application concerns electrical networks, where potential/current or voltage/current variables form conjugate pairs. Interconnecting ports gives linear constraints and suitable ideal components admit a Lagrangian-relational semantics. To describe resistors, sources or dynamical components, the chosen coefficient field and the individual constitutive laws still matter. The diagrams do not magically make every physical component symplectic or lossless.

This also clarifies the family resemblance to ZX-calculus. Both use coloured generators, spiders and equations that permit local replacements. The intended semantic objects are nevertheless different:

$$\begin{aligned} \mathbf{cb}_k &\longleftrightarrow \text{matrices}, \\ \mathbf{ih}_k &\longleftrightarrow \text{linear relations}, \\ \mathsf{LagRel}_k &\longleftrightarrow \text{Lagrangian phase-space relations}. \end{aligned}$$

A node is given its mathematical meaning by its interpretation, not by its colour or visual resemblance to some other diagram.

## §1.17 Returning to the first network

The original weighted network represented

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

In $\mathbf{cb}_k$, it can be compared with any other weighted copy–scalar–add circuit using a sound and complete equational calculus. Viewed in $\mathsf{LinRel}_k$, its graph can be reversed, connected with a zero test, or placed inside feedback; the resulting system may no longer be a function. Finally, as a doubled relation in phase space it carries the pair of constraints

$$Q = Aq, \qquad p = A^{\mathsf{T}} P.$$

The transpose and the reversed momentum wire now come from a mathematical requirement: compatibility with the symplectic pairing.

At each stage the pictures require a precise rule for what their boundaries mean and for what counts as equality. The diagrammatic method earns its value when it permits us to manipulate a network while preserving its linear or symplectic semantics; a visually pleasing picture without those rules would be little more than an illustration.